PAPERGAMES

Towers of Hanoi

Patterns and counting

Open the game

Name: _____

Goal: You find a number pattern and learn to split a big task into small ones.

Rules: All discs are on the left rod A, the biggest at the bottom. Move the whole tower to the right rod C. You move one disc at a time, and a bigger disc never lies on a smaller one.

1. How many moves do you need at least for 1, 2 and 3 discs? Try it with coins or pieces of paper.

2. Fill in the table. What pattern do you see?

Discs	1	2	3	4	5
Moves					

3. How many moves do you need at least for 8 discs?

4. Write down all 7 moves for 3 discs from A to C (for example: small disc from A to C).

Free from PaperGames. You may copy it freely for lessons. https://yourchaoschris.github.io/PaperGames/

PAPERGAMES

Solutions for teachers

Towers of Hanoi

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1. 1 disc: 1 move, 2 discs: 3 moves, 3 discs: 7 moves.
2. 1, 3, 7, 15, 31. Each number is double the one before plus 1. As a formula: $2^n - 1$.
3. 255 moves.
4. 1. small A→C, 2. middle A→B, 3. small C→B, 4. big A→C, 5. small B→A, 6. middle B→C, 7. small A→C.

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